Chapter 3.3 — Formula Sheet

Selection Bias, Adverse Selection & Information Economics

Companion quick-reference to your Notebook 3 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

A. Conditioning on Trade Execution

Core Identity [Ex 1–2]

$$\mathbb{E}[V \mid \text{Trade Executed}] \neq \mathbb{E}[V]$$

A resting order only fills when the counterparty wants to trade — that want is itself information. Ask fill ($V >$ ask) and bid fill ($V <$ bid) are mirror images of the same truncation idea.

Truncated-Uniform Shortcut [Ex 1–3]

For $V \sim \text{Unif}(L, M)$:

$$\mathbb{E}[V \mid V > \text{ask}] = \frac{\text{ask} + M}{2}, \qquad \mathbb{E}[V \mid V < \text{bid}] = \frac{L + \text{bid}}{2}$$

Truncating a uniform always lands at the midpoint of the *surviving* range — continuous or discrete, the logic is identical (Ex 3).

Giveaway & Capping the Loss [Ex 4–5]

$$\text{giveaway} = \mathbb{E}[V \mid \text{fill}] - \text{quote}$$

Setting giveaway = a target cap and solving for the quote is just inverting the truncated-mean formula above — same identity, read backwards.

B. The Winner's Curse

Baseline: Signal Alone, No Competition [Ex 6]

$V \in \{L, H\}$ equally likely, signal correct w.p. a:

$$\mathbb{P}(V{=}H \mid S{=}H) = a \quad \Rightarrow \quad \mathbb{E}[V \mid S{=}H] = L + (H - L)a$$

Ordinary Bayesian updating — nothing auction-specific yet.

The Curse, Fully Derived [Ex 7–8]

2 bidders, signals i.i.d. given V. Win (bid = signal, ties coin-flipped) iff $S_1 \geq S_2$:

$$\mathbb{P}(S_2{=}L \mid S_1{=}H) = 2a(1-a), \quad \mathbb{P}(S_2{=}H \mid S_1{=}H) = 1 - 2a(1-a)$$

$$\mathbb{E}[V \mid S_1{=}H, S_2{=}L] = \frac{H + L}{2} \text{ (signals cancel — always the prior!)}$$

$$\mathbb{P}(\text{win} \mid S_1{=}H) = 2a(1-a)(1) + \big(1 - 2a(1-a)\big)\left(\tfrac{1}{2}\right)$$

Then, with $w_L = \mathbb{P}(S_2{=}L \mid S_1{=}H)$, $w_H = \mathbb{P}(S_2{=}H \mid S_1{=}H)$:

$$\mathbb{E}[V \mid S_1{=}H, \text{win}] = \frac{w_L \cdot \frac{H+L}{2} + w_H \cdot \frac{1}{2} \cdot \mathbb{E}[V \mid S_1{=}S_2{=}H]}{\mathbb{P}(\text{win} \mid S_1{=}H)}$$

Rational bid $= \mathbb{E}[V \mid S_1, \text{win}]$, strictly below both the raw signal and the no-competition estimate.

Signal Quality & Bidder Count [Ex 9, 11]

More bidders ⇒ deeper curse (beating more rivals is stronger evidence). Curse gap vs. a: **zero at** $a = 0.5$ (uninformative signal, Ex 5 of this drill), grows through intermediate a, then shrinks back toward 0 as $a \to 1$ (signal becomes definitive, nothing left to correct).

C. Trading Against Asymmetric Information

Glosten–Milgrom, Symmetric 50/50 Prior [Ex 12–14]

Fraction α informed (buys iff $V{=}H$, sells iff $V{=}L$); rest noise (50/50 regardless of V):

$$\mathbb{P}(\text{buy} \mid H) = \alpha + \tfrac{1-\alpha}{2}, \qquad \mathbb{P}(\text{buy} \mid L) = \tfrac{1-\alpha}{2}$$

Closed form (50/50 prior only):

$$\text{ask} = \text{FV} + \alpha \cdot \frac{H - L}{2}, \qquad \text{bid} = \text{FV} - \alpha \cdot \frac{H - L}{2}$$

$$\text{spread} = \alpha(H - L) \quad \text{— exactly linear in } \alpha$$

Reading α Backwards; Asymmetric Priors [Ex 15–16]

$$\alpha = \frac{\text{spread}}{H - L} \quad \text{or} \quad \alpha = \frac{\text{ask} - \text{FV}}{(H - L)/2}$$

When $p = \mathbb{P}(V{=}H) \neq 0.5$, the clean formula breaks — redo full Bayes on *each side separately* (ask-side and bid-side info half-spreads are no longer equal).

D1. General Glosten–Milgrom Formula [Ex 17]

For general prior $p = \mathbb{P}(V{=}H)$:

$$\mathbb{P}(\text{buy} \mid H) = \frac{1 + \alpha}{2}, \qquad \mathbb{P}(\text{buy} \mid L) = \frac{1 - \alpha}{2}$$

$$\mathbb{P}(H \mid \text{buy}) = \frac{p(1 + \alpha)}{p(1 + \alpha) + (1 - p)(1 - \alpha)}$$

$$\text{ask} = H \cdot \mathbb{P}(H \mid \text{buy}) + L \cdot \big(1 - \mathbb{P}(H \mid \text{buy})\big)$$

$p = 0.5$ collapses this to Section C's linear form; any other p needs the full fraction.

D2. Sequential Updating [Ex 18]

Feed each trade's posterior in as the *next* trade's prior. A buy then a sell (equal α each side) **exactly cancels** back to the original prior — order flow is a reversible running Bayesian update, not a one-way ratchet.

D3. Stacking Information Risk on Inventory Risk [Ex 19–20]

Two *independent, additive* skews — compute separately, then combine:

$$r = \text{FV} - q\gamma\sigma^2 \quad \text{(Ch. 3.2 — shifts the } \textit{center}\text{)}$$

$$\text{bid} = r - (\text{info half-spread}), \qquad \text{ask} = r + (\text{info half-spread})$$

Symmetric prior: info half-spread $= \alpha \cdot \frac{H-L}{2}$ on both sides. Asymmetric prior: use each side's *own* info half-spread (ask-side ≠ bid-side) — never a shared average.

One-tool-tree reminder: every fill, every auction win, and every informed order is the same event in disguise — a counterparty choosing to trade with you because they know something you don't. Section A names it, Section B prices it under competition, Section C defends a two-sided quote against it, and Section D stacks that defense on top of Ch. 3.2's inventory logic. One Bayesian update, four settings.